

L9: SI Model

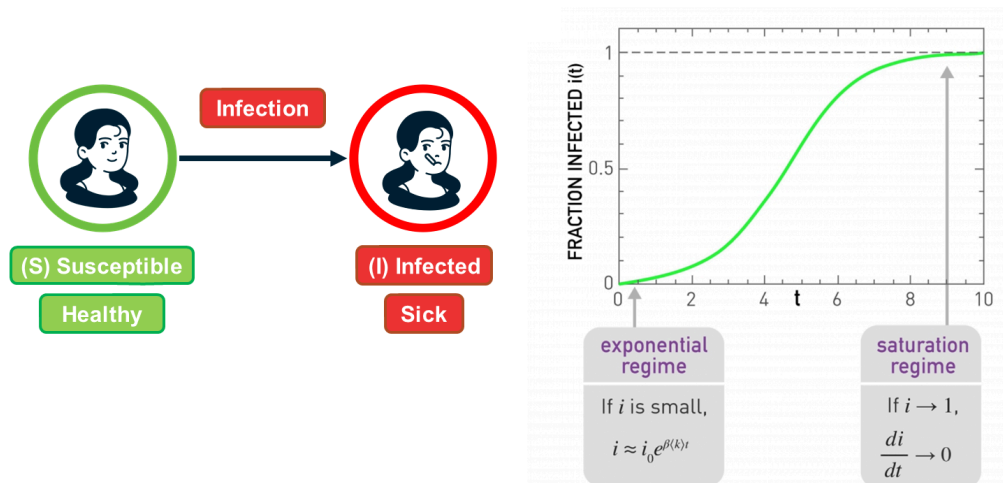


Figure 10.5 from Network Science by Albert-László Barabási

Suppose that we have N individuals in the population and, according to the homogeneous mixing assumption, each individual has the same number of contacts $\bar{k}$ (this is shown as $\langle k \rangle$ in the textbook visualizations).

In the SI model, there are two compartments: Susceptible (S) and Infected (I) individuals. To become infected, a susceptible individual must come in contact with an infected individual. If someone gets infected, they stay infected.

If $S(t)$ and $I(t)$ are the number of susceptible and infected individuals as functions of time, respectively, we have that $S(t) + I(t) = N$. We typically normalize these two functions by the population size, and we work with the two population densities: $s(t) = S(t)/N$ and $i(t) = I(t)/N$, with $s(t) + i(t) = 1$.

The infection starts at time=0 with a single infected individual: $i(0) = 1/N = i_0$.

Suppose that an S individual is in contact with only one infected individual. Let us define the parameter β as follows: βdt is the probability that S will become infected during an infinitesimal time interval of length dt .

Given that the S individual is in contact with $\bar{k}$ infected individuals, this probability increases to $1 - (1 - \beta dt)^{\bar{k}} \approx 1 - (1 - \bar{k}\beta dt) = \bar{k}\beta dt$ because the infection can take place independently through any of the infected contacts (the approximation is good as long as this probability is very small).

If the density of infected individuals is $i(t)$, then the probability that the S individual becomes infected is $\bar{k}\beta i(t) dt$.

The infection process is always probabilistic but for simplicity, we can model it deterministically with a two-state continuous-time Markov process: an individual moves from the S to the I state with a transition rate $\bar{k}\beta i(t)$.

So, if the density of S individuals is $s(t)$, the increase in the density of infected individuals during dt is:

$$di(t) = \bar{k}\beta i(t) s(t) dt$$

Thus the SI model can be described with the differential equation:

$$\frac{di(t)}{dt} = \bar{k}\beta i(t) (1 - i(t))$$

with initial condition $i(0) = i_0$.

This is a nonlinear differential equation (because of the quadratic term) but it can be solved noting that

$$\frac{di}{i(1-i)} = \frac{di}{i} + \frac{di}{1-i} = \bar{k}\beta dt$$

where we replaced $i(t)$ with i for simplicity.

Integrating both sides, we get that:

$$\ln i - \ln(1-i) = \bar{k}\beta t + \text{constant}$$

The initial condition gives us that this constant is equal to $\ln i_0 - \ln(1-i_0)$.

So, if we exponentiate both sides of the previous equation we get that:

$$\frac{i}{1-i} = \frac{i_0}{1-i_0} e^{\bar{k}\beta t}$$

and so we get the closed-form solution for the density of infected individuals for $t \geq 0$:

$$i(t) = \frac{i_0 e^{\bar{k}\beta t}}{(1-i_0) + i_0 e^{\bar{k}\beta t}} \quad (1)$$

This function is plotted at the visualization.

There are some important things to note about this function:

1. For small values of t , when the density of infected individuals is very small and the outbreak is only at its start, $i(t)$ increases exponentially fast: $i(t) \approx i_0 e^{\bar{k}\beta t}$
2. The time constant during that “exponential regime” is $\frac{1}{\bar{k}\beta}$. This time constant is often used to quantify how fast an outbreak spreads. This time constant decreases with both the infectiousness of the pathogen (quantified by β) and the number of contacts $\bar{k}$.
3. For large values of t , the density of infected individuals tends asymptotically to 1 – meaning that everyone gets infected.

Food For Thought

Perform the last derivation, showing how to get equation (1), in more detail.